Lightsheet Analysis Protocol(ish): A Guide to Avoiding all the Mistakes EGV Made
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Section 1: Pipeline Processing Prep

All associated code can be found at: https://github.com/emily-gv/LSFM-Pipeline-Mar2026.git
Code can be run locally to test that everything is working, but batch analysis should be done using a compute cluster. Some steps can take 30hrs or upwards of 750GB RAM to run and your computer will be incapacitated.

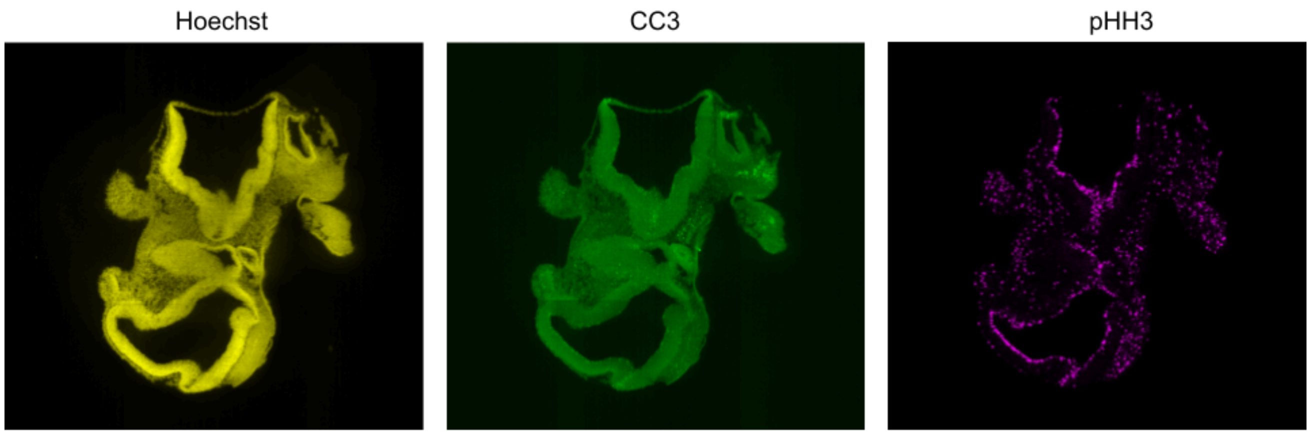
You can run the pipeline in phases by commenting/uncommenting samples to be worked with & flags to use in the configuration file (further explained below).

Prerequisites:
- A computer running **Linux**
- Git
    - If you don't have already it, run in terminal: `sudo apt install git`
- Anaconda/miniconda: https://www.anaconda.com/download
- Models to segment your cell markers of choice
    - Currently available models: tissue segmentation (mesenchyme + NE), nuclear (Hoechst & CSGreen validated - high intensity LSFM images), pHH3, CC3
    - To train your own model, see **Section 6**

Prep: Clear and stain samples following the lab's modified CUBIC protocol

Part 1: Downloading from the lightsheet

1. Manually adjust the channel intensities to minimize colour in the background while retaining the whole sample
    a. Note: the currently trained CC3 model tolerates background (see examples below)



2. Download your files as a z-stack
    - I kept them on the NAS, and then uploaded them to the server (or download locally) when ready to work with the compressed versions

    On Zen Blue, select 'Method: Image Export' with the following parameters
    - File Type: 'Tagged Image File Format (TIFF)'
    - NO CLICK: 'Burn-in Annotations'
    - CLICK: 'Short Format'
    - NO CLICK: 'Use Channel Names'

Part 2: Preparing your environment

1. Clone the github repo
    - https://docs.github.com/en/repositories/creating-and-managing-repositories/cloning-a-repository
2. In terminal, navigate to the folder 'LSFM-Pipeline-Mar2026'
3. Run:
    ```
    conda env create -f lsfm_pipeline.yml
    ```
    - Activate the environment using: `conda activate lsfm-pipeline`
4. With the environment activated, run:
    ```
    conda install -c nvidia cuda-toolkit=12.2
    conda install -c conda-forge libxcb
    ```

You are now ready to begin processing the tiff stacks.

Part 3a: Setting up your configuration file (config.yml) - adding a new cell marker (config.yml)

1. Under the 'markers' section (line 42), add a new entry corresponding to your marker. It should look like:

   - name: 'marker-name'
     window_cellpose: 1024 (model-specific, see **Section 6**)
     flag_compress: False
     flag_segment: False
     flag_mesenchyme: False
     flag_remove_sedimentation: False

2. Uncomment and copy everything in the section at the top and paste it at the bottom of the file (or go line by line and paste your new versions under with their cc3/phh3 counterpart lines if you want to keep it clean and nice)
   - IMPORTANT: when replacing the marker name ensure capitalization is consistent with how you wrote it in Step 1

- folder_marker-name_tiles: "Step11_marker-name_tiles/Step12_marker-name_tiles"
- folder_marker-name_segmented_tiles_label: "Step22_marker-name_tiles_label"
- folder_marker-name_segmented_tiles_binary: "Step22_marker-name_tiles_binary"
- folder_marker-name_slices_label: "Step25b_marker-name_slices_label"
- folder_marker-name_slices_binary: "Step25b_marker-name_slices_binary"
- folder_marker-name_masked_label: "Step25b_marker-name_masked_label"
- folder_marker-name_masked_binary: "Step25b_marker-name_masked_binary"
- dest_file_marker-name_label_tiff: "_Step26_marker-name_label_raw.tiff"
- dest_file_marker-name_binary_tiff: "_Step26_marker-name_binary_raw.tiff"
- dest_file_marker-name_label_tiff_maskout_sedimentation: "_Step33_marker-name_label.tiff"
- dest_file_marker-name_binary_tiff_maskout_sedimentation: "_Step33_marker-name_binary.tiff"

3. Go to file AuxFunctions/sort_files.py. Add a comma on line 28 and uncomment line 29, adding your marker name (ensure capitalization is consistent with before)

Part 3b: Setting up your configuration file (config.yml)

1. Fill in 'samples' with your sample details
   - folder_slices: full path to the folder of slices (can be NAS)
   - resX/Y/Z should be taken from Zen Black (Zen Blue doesn't give the precise resolution). Open your file and go to "Info"
   - Add a new line if necessary: subfolder_new-marker

2. Fill in the following with the full filepaths (stored locally):
   - folder_output
   - folder_CNN_architectures

3. Fill in the following with the file names: (add your new cell marker if necessary)
    - filename_CNN_tissues
    - filename_CNN_nuclei
    - filename_CNN_cc3

Section 2: File Preprocessing

** Make sure you have the conda environment activated**

Part 1: Image sorting & compression

1. Go to AuxFunctions/sort_files.py and check that the c1/c2/c3.tif (under output_folders, line 25) correspond to the correct channels/markers from the lightsheet
2. In config.yml, for all the marker channels you want to compress under 'markers' set 'flag_compress: True'
    - If you download a new sample everything might be set to True, but this way you can save time if you go back and adjust/re-download a single channel
3. In terminal, run: `python sort_and_compress.py`

After this step you are ready to begin segmenting the images.

Section 3: Segmentations

** Make sure you have the conda environment activated**

Part 1: Tissue Segmentation

Try following the steps to run as-is. If it crashes severely, try uncommenting and editing the following lines in main_01_tissue_segmentation.py:

- os.environ['XLA_FLAGS']
- os.environ['PATH']
- os.environ['LD_LIBRARY_PATH']

1. Ensure you have samples commented/uncommented in config.yml as desired
2. In terminal, run: `python main_01_tissue_segmentation.py`

Part 2: Cell Segmentation

1. For each marker you want to segment, toggle 'flag-segment: True' under 'markers' in config.yml
2. If you are interested in tracking cells in the edge of the mesenchyme, toggle 'flag_mesenchyme: True'
3. In terminal, run: `python main_02_tissue_segmentation.py`

By the end of this section you should have:

a. TIFF volumes of the mesenchyme, NE, and mixed tissue

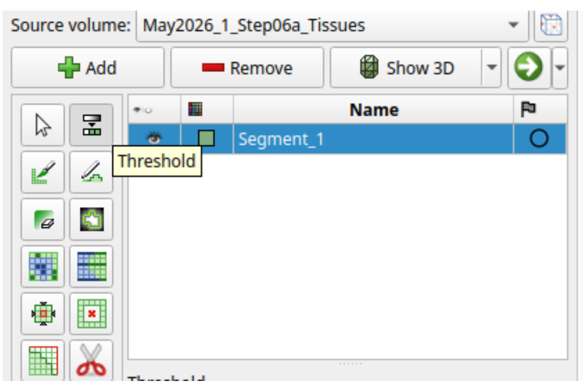
b. GAN-related TIFF volumes
c. TIFF volumes of your markers of choice (binary and label versions)
    - Binary is just your black and white segmentations
    - Label is Cellpose's attempt at also categorizing # of cells and other things
        - It's not too accurate in my experience. If you want to identify individual cells in the image I'd recommend using SciPy's label function (`from scipy.ndimage import label`)

Section 4: Post-segmentation processing

Prerequisites:
- 3DSlicer
- Paraview (recommended for visualization)
- XNView Multi Platform (recommended for visualization)

Part 1: Removing sedimentation

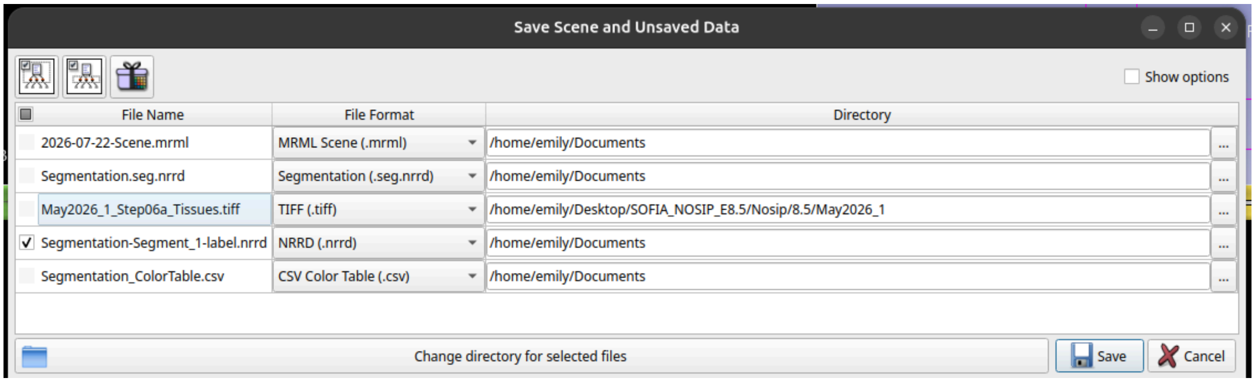
1. Begin by opening the file '_Step6a_Tissues.tiff' for one of your segmented samples
2. Navigate to the Segment Editor module, then click 'Add' to add a new segmentation
3. Click the Threshold widget, then adjust the Threshold Range so that all tissues flash green (min 40-100 max works)

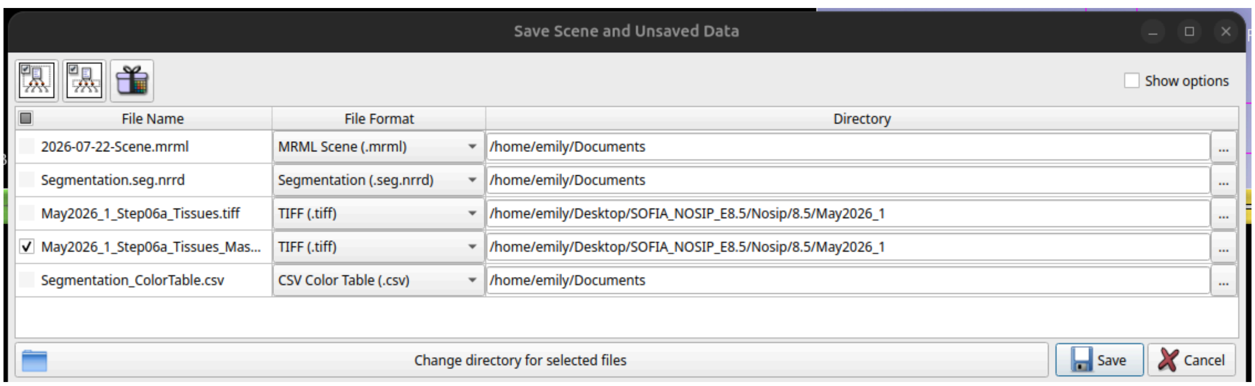


4. Click 'Show3D' and center the image using the blue box buttons. Enlarge the blue box and you can use the scissor tool to cut around and remove any sedimentation.
5. For more precise correction, use the eraser tool on one of the red/green/yellow box images (be sure to turn off Show3D before you do this for it not to lag)

<u>Help, my computer keeps crashing!</u>
The TIFF likely has a smaller z step size, so it's a significantly larger file and uses up all your RAM. In addition to adding extra swap space, you can:
Open 3DSlicer from scratch and navigate to Segment Editor. Open python console with CTRL + 3, then type the following command:
slicer.modules.SegmentEditorWidget.editor.setMaximumNumberOfUndoStates(2)

6. Once finished with corrections, navigate to the module ‘Data’ and right click on the line ‘Segment_1’. Select ‘export visible segments to binary labelmap’
7. Click the ‘Save’ icon at the top of the screen. Update the name/file extension & path for the .nrrd file. Rename it to **<sample-name>_Step06a_Tissues_Mask.tiff.** The file format drop down should adjust automatically. Save only that file.





8. For all markers you’re working with and have TIFF files for, toggle ‘flag_remove_sedimentation: True’ in the config.yml file
9. In terminal, run `python main_03_remove_sedimentation.py`
    - If you want to re-run this for just a cell marker and NOT the tissue files, you can toggle ‘flag_tissue = False’ in the main_03 file

Part 2: Move NE to Mesenchyme

Section 6: Using the pipeline on ARC

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1. Connect to ARC via SSH. In terminal, type: ssh <login.lastname>@arc.ucalgary.ca

- If you aren’t on campus wifi you will need a VPN: https://rcs.ucalgary.ca/Connecting_to_RCS_HPC_Systems

2. In the terminal (not on ARC), upload the code folder using the scp command. For example: scp -r /home/emily/Documents/LSFM-Pipeline-Mar2026/script_register_to_reference.py emily.garciavolk@arc.ucalgary.ca:/home/emily.garciavolk/

    - Replace the first part with your local filepath to the folder, and the second replace the name(s) with your login credentials

3. Follow this guide to set-up conda on ARC: https://rcs.ucalgary.ca/Conda_on_ARC#Creating_Conda_environments. Use the file tensorflow-20250512.yaml. If that doesn't work, try environment-tensorflow-cellpose-arc.yml.

4. Within the LFSM-Pipeline-Mar2026 folder, create a subfolder called 'ARC'. This will house your SLURM scripts. Anytime you want to run a script, run ARC/xyz.slurm (the one in the repo comes set up with the ARC format for including conda in a job) from the LFSM-Pipeline-Mar2026 folder.
    - I recommend running one sample at a time to test out how time/storage requirements compare to the limits on your server. For each job you submit, you will need to copy the files you use and adjust the files accordingly (depends, but I believe if you run script1.py using config1.yml and then adjust config1, if it's in queue script1 will end up running using the edited file. Save yourself the hassle of finding out by using separate files per sample).
    - For example:

        main_01_tissue_segmentation_May2026_105_1.py
        main_01_tissue_segmentation_May2026_105_2.py
        config_May2026_105_1.yml
        config_May2026_105_2.yml
        main_02_cell_segmentation_May2026_1.py
        main_02_cell_segmentation_May2026_2.py

If you're not at UCalgary, find your respective university's server guide and set up the conda environment following those instructions.